

2023

COMPUTER SCIENCE AND ENGINEERING

Paper : CSEL-1013

(Computational Geometry)

Full Marks : 70

The figures in the margin indicate full marks.

Candidates are required to give their answers in their own words as far as practicable.

Answer **question no. 1**, **question no. 2**, and **any four** from the rest.

1. Define and explain **any five** from the following : 2×5
 - (a) Scan reflex vertex
 - (b) Element uniqueness problem
 - (c) Order reversing
 - (d) Left-hand turn
 - (e) x-Monotonicity of a polygon
 - (f) Nearest neighbour problem
 - (g) Maximizing angles and edge flipping.

2. Answer **any five** from the following : 4×5
 - (a) What do you mean by *inversion counting*? For a given set of pairs of points $\{(a_i, b_i), (a_j, b_j)\}$, explain how inversion counting may help to count negative slopes.
 - (b) Define and explain the term *fix-up* related to the polygon triangulation problem, and show how it plays its role to triangulate a monotone polygon with necessary diagrams.
 - (c) Write a short note on *discrete combinatorial geometry* in the context of its theory and applications in order to solve geometric computational problems.
 - (d) Define the *half-plane intersection problem*. Outline the divide-and-conquer based algorithm for solving the same and compute its time complexity.
 - (e) Define and explain Voronoi diagram with the help of a suitable example. Clearly explain how such a diagram can be utilized to compute the convex hull for a given set P of n sites on the plane.
 - (f) What do you mean by *Delaunay triangulation*? Briefly discuss *spanner property*. Is there any relation between Delaunay triangulation and spanner property for a set of points given on a plane?
 - (g) Define *upper envelope* and *lower envelope* related to half-plane intersection problem. How these two may help to compute convex hull following the theory of point-line duality?

Please Turn Over

3. (a) Define and explain convex hull.
 (b) Critically compare the underlying algorithms in brief along with the complexity issues to compute convex hull by the methods of Graham's scan and Jarvis's march. 2+8
4. (a) What do you mean by *Randomized Incremental Algorithm* <RIA>?
 (b) Clearly state the assumptions, if any, and show how such an RI algorithm could be devised in computing trapezoidal maps for a given set of line segments.
 (c) In this context, if the number of line segments is n , then what are the maximum number of vertices and the number of trapezoids obtained in the resulting trapezoidal map, and prove them. 2+4+4
5. (a) Define and explain the following terms with suitable diagrams:
 (i) Voronoi complex, (ii) Voronoi edge, (iii) Voronoi vertex and its degree.
 (b) State how and where the beach line is formed, and explain its properties and progress in computing Voronoi diagram following Fortune's algorithm. $(1\frac{1}{2}+1\frac{1}{2}+2)+5$
6. (a) Explicate the following statement with necessary diagrams: "The minimum spanning tree of a set of points is a subgraph of the Delaunay triangulation of the points." Is it true in general?
 (b) Consider the following three line equations in the primal plane :

$$l_1 : y = 2x + 1; l_2 : y = -x/2 + 6; l_3 : y = 5(x + 1)/3.$$
 Comment about the points l_1^* , l_2^* , and l_3^* in the dual plane, and prove it. $(4+1)+5$
7. (a) Solve the problem of *slope range counting*, where given a set S of n non-vertical lines in a real two-dimensional plane and also given an interval $[s^-, s^+]$, the objective is to count the pairs of lines whose intersections lie within the vertical slab whose left side is $x = s^-$ and right side is $x = s^+$. Deduce its time complexity.
 (b) Define the *line segment intersection problem* and discuss the data structures while executing an algorithm to find the events and detect intersections. $(4+2)+4$
8. (a) State and explain the *multi-dimensional constrained optimization problem*, and show how this problem is geometrically related to bounded, infeasible, or unbounded polytopes.
 (b) Briefly describe an algorithm to compute the intersectional convex polygon when a pair of convex polygons intersect to each other, and also quantify its time complexity. $(2+3)+(4+1)$